	Test on Ground-Truth from Model A AUROC / Brier Score	Test on Ground-Truth from Model B AUROC / Brier Score
Starting from Model A	0.8633 / 0.1491	0.8460 / 0.1582
Starting from Model B	0.8631 / 0.1497	0.8717 / 0.1443

Table 3 'Cross-Experiments': We trained P(IK) classifiers on the ground-truth data from two models, starting from both models. Ideally, starting from pretrained model X should do better than starting from model Y when training P(IK) using data from model X. We see some signal that starting from model B does better than starting from model A when testing on data from model B. However, we see no difference between both initializations when testing on data from model A.

6 Discussion

The long-term motivation underlying this work is to begin to understand aspects of *honesty* in AI models. We use honesty as an umbrella term including several overlapping[7] ideas:

- Truthfulness: Does the AI provide factually accurate information, including finding, using, and evaluating source materials correctly?
- Calibration: Do the AI's probabilistic predictions correspond with frequencies of occurrence?
- Self-knowledge: Do AI systems know what they know and make accurate predictions about their own behavior and reasoning?
- Explainability: Do AI systems reveal their 'thinking' completely and faithfully?
- Non-deceptiveness: Can we ensure AI systems do not learn to lie (e.g. when human preference data might encourage systematic mistakes or provide more reward for pleasant misconceptions)?

Here we have focused on aspects of calibration, self-knowledge, and truthfulness, but we are only scratching the surface of the larger subject. Our core findings are that large language models can be well-calibrated on diverse multiple choice questions, that they perform well at self-evaluation on a range of subjects, and that they can be trained to predict what they do and don't know and exhibit some generalization to new domains and in-context information sources.

We found that calibration tends to improve with model size/capability and with additional few-shot examples. Self-evaluation also improves with model size, which is non-trivial since we expect the quality of model samples to also be improving as we scale up. This means that in effect, we are observing that verification has an advantage over generation. Within the scope of pretrained language models, we find these results encouraging, and expect they can be usefully leveraged to bootstrap more extensive forms of self or inter-model supervision, and to distill calibration back into open-ended natural language responses.

Another motivation for this work was to use self-knowledge as a test-bed for generalization. We are particularly interested in how honesty generalizes from easy to hard questions and across domains, because we are worried about how AI systems may behave out-of-distribution, or when they have some knowledge or insight that humans lack. We found that language models do exhibit a degree of generalization across domains, though calibration suffers out-of-distribution. We also found that although models are trained to predict P(IK) for standalone questions, they generalize in such a way that P(IK) increases significantly when models are provided with source materials that contain the answers to these questions, or hints towards the solution of math word problems. This suggests a relationship between the way models store longterm memory and the way they use information within their contexts, i.e. it suggests a connection between learning and in-context learning.

6.1 Limitations and Future Work

This work has a number of limitations:

[7] We are not attempting a precise taxonomy, but simply pointing at a collection of important ideas.

- We have focused on pretrained language models, but models that have been finetuned for a specific purpose, such as RLHF policies [Bai et al., 2022], will not be so well-calibrated, and some of the methods we discussed may not work as well.
- Furthermore, language model pretraining is a form of human imitation. Some of the most interesting questions about honesty concern setups where advanced AI systems 'know' something that clashes with or extends what humans know, and so cannot have been grasped from pretraining. Our analysis does not differentiate between "the truth" and "what humans say", but eventually we would like to train AI systems that recognize this distinction and generalize appropriately.
- This work does not address the possibility that AI systems may learn to behave deceptively (in order to e.g. maximize human preference rewards), and intentional deception would counteract simple trends in honest self-evaluation.
- We were interested in the generalization of honesty because we would like to predict whether AI systems will extrapolate from training to exhibit safe and desirable behavior in new contexts. We found some interesting and potentially encouraging forms of generalization. But we only studied five sampling-based datasets, and our observations concerning capability trends and generalization were limited in power and scope. It would be interesting to investigate these more thoroughly and precisely.

With that said, we believe that the simple methods we have demonstrated here lay the foundation for investigating some of these issues in the future.

6.2 Broader Impacts

We hope that the broader impacts of this work will be positive. Pretrained language models and other AI systems can be dishonest in a number of ways, and we hope that this work will make it easier to identify and correct failures of honesty in deployed systems. That said, pretrained language models may have other failure modes, and the interventions we study here may not correct errors that are perpetuated by language model training data and human imitation.

7 Contribution Statement

Model Training: Model pretraining was led by Sam McCandlish, Nicholas Joseph, Tom Brown, and Jared Kaplan, and the majority of Anthropic's technical staff contributed to the development of our efficient distributed training infrastructure and the underlying machine learning systems. Core contributors include Tom Henighan, Scott Johnston, Sheer El Showk, Nicholas Joseph, Nelson Elhage, and Ben Mann. Scott Johnston and Sheer El-Showk in particular worked on optimizing pretraining for ML efficiency.

Sampling and Evaluation: Efficient sampling efforts were led by Tom Brown, and Tom Conerly carried out major aspects of the design, implementation and support for the system, with help from Zac Hatfield-Dodds. Many members of Anthropic worked on evaluations, including Saurav Kadavath, Nicholas Schiefer, Nick Joseph, Tom Henighan, Amanda Askell, Jared Kaplan, Andy Jones, and Sam McCandlish.

Cluster: Nova DasSarma and Eli Tran-Johnson managed the research cluster our research depended on and maintained its stability, making this research possible. Many others helped with these efforts, including Ben Mann, Tom Henighan, Sam McCandlish, Andy Jones, and Tristan Hume.

Research: Saurav Kadavath and Jared Kaplan designed and carried out most of the experiments. Saurav built much of the relevant infrastructure for the experiments, including for P(IK) training and large scale evaluations. Amanda Askell and Ethan Perez helped with experiments and design. Tom Henighan carried out early experiments surveying a variety of evaluation formats. Dawn Drain helped set up the code evaluations, and obtained the python function synthesis examples with test coverage. Anna Chen did early work training models to identify relevant source materials.

Writing: This paper was drafted by Jared Kaplan and Saurav Kadavath. Other members of Anthropic made miscellaneous contributions and suggestions throughout the writing process.

Other contributions: The ideas explored in this paper developed in conversations with many of Anthropic's staff, especially Yuntao Bai, Stan Fort, Deep Ganguli, Jackson Kernion, Dario Amodei, Catherine Olsson, Sam Bowman, Sam McCandlish, and Chris Olah.

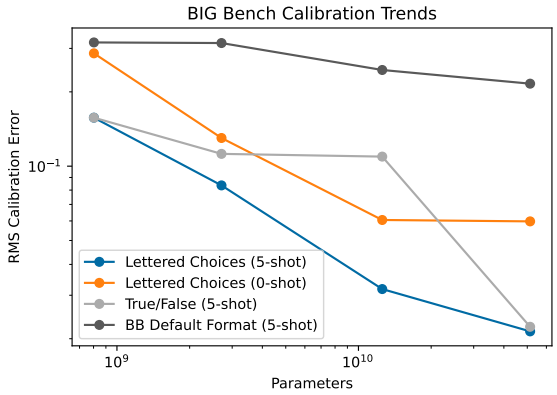


Figure 21 Here we show trends in the RMS calibration error on BIG Bench, for both multiple choice and a separate True/False format. It's qualitatively very similar to the ECE pictured in Figure 4

Acknowledgments

We thank Paul Christiano for helpful discussions, and Owain Evans, Roger Grosse, Dan Hendrycks, Jacob Hilton, Geoffrey Irving, and Daniel Ziegler for comments on the draft. We're also deeply grateful to Daniela Amodei, Jarrah Bloomfield, Jamie Kerr, Timothy Telleen-Lawton, Jia Yuan Loke, Jeffrey Ladish, Rebecca Raible, Rune Kvist, Rob Gilson, Guro Khundadze, Filipe Dobreira, and Sebastian Conybeare for their help and support.

A Metrics, Formatting Details, and P(IK) Training

A.1 Calibration Charts

To evaluate calibration on multiple choice problems, we obtain a list of all probabilities for all answer options (both correct and incorrect choices). We then sort these probabilities and bin them into 10 groups, each with an equal number of probabilities. We obtain the x-coordinate of calibration charts by computing the mean probability in each of these bins. To obtain the y-coordinate, we compute the fraction of the options that actually correspond to a correct answer within each bin. The result is a calibration chart such as Figure 6. Our method differs slightly from some others in the literature [Guo et al., 2017, Srivastava et al., 2022, Lin et al., 2022], where only the probabilities for the top prediction are included.

A.2 Expected Calibration Error

To compute the ECE, we simply evaluate

$$E = \frac{1}{N}\sum_{i}^{N} |y_i - x_i| \tag{A.1}$$

for the (x_i, y_i) coordinates of our calibration charts (which have equal numbers of predictions in each bin). This results in plots like Figure 4. Note however that when computing the ECE we only use the top predictions for each multiple choice question, as this seems to be the convention in the literature (this change results in a larger ECE, compared to if we had used all predictions).

We also evaluate the RMS Calibration Error

$$E = \sqrt{\frac{1}{N}\sum_{i}^{N} (y_i - x_i)^2} \tag{A.2}$$

which appears less often in the literature, though it has a sounder theoretical basis.